

TUTORIAL TIP SYSTEM

OVERVIEW

The Tutorial Tip System is responsible for managing the light-bulb alerts that appear in the upper left and do their best to help out the player. The primary script used in this system is `TutorialTipManager.cs`. It sits on the `TutorialTipManager` game object. Also under the Tutorial Tip Manager game object is an object named `TutorialTipTextObjects` which holds a bunch of text objects, one for each possible tutorial tip. Another important script involved in this system is `TutorialTip.cs`, which is an abstract class from which all the Tutorial Tip types inherit.

Essentially, the way the system works is that for every tutorial tip that can appear, there is an associated class that inherits from `TutorialTip`. Each of these child classes is placed on the text object it should display (all of which are under the `TutorialTipTextObjects` game object). At the start of the game, each tutorial tip calls its `InitializeThisTutorialTip()` function, which adds listeners for the events that should activate (and deactivate) the tip. When the events that could activate the tip occur, the tutorial tip checks to see if all of its activation conditions have been met. If they have been, it tells the tutorial tip manager to activate it.

When a tip tells the tutorial tip manager to activate it, the tutorial tip manager adds it to the bottom of its list of active tutorial tips. If multiple tips are active at the same time, it will only display the first one and then go down the list once that one is deactivated. (Side note: It can either be deactivated because its deactivation event is met, or because the player closed it.)

HOW TO ADD A TUTORIAL TIP

To create a new tutorial tip, you basically just have to create a new tutorial tip text object and then create an associated class to activate it that inherits from TutorialTip. Here are the steps:

1. Create a new TextMeshPro text object under TutorialTipManager -> TutorialTipTextObjects.
 - a. In the inspector, go to the TutorialTipManager class and add the new text object you created to the tutorialTipTextObjects array
 - b. In the inspector, under the TutorialTipManager class, set the time (in seconds) that should be waited before activating and deactivating the tutorial tip after activation/deactivation conditions are met. (Use the timesToWaitBeforeActivating and timesToWaitBeforeDeactivating arrays)
2. Create a child class of the TutorialTip class for the tutorial tip you're making and attach it to the text object you created in step one.
 - a. Place the new script you made in this folder:
Assets/Scripts/UIScripts/TutorialTips/TutorialTipTypes
 - b. Add this constructor to the class, replacing ExampleTutorialTip with the name of the class: `“public ExampleTutorialTip(int _tutorialTipTextID, TutorialTipManager _TTM, float _timeToWaitBeforeActivating, float _timeToWaitBeforeDeactivating) : base(_tutorialTipTextID, _TTM, _timeToWaitBeforeActivating, _timeToWaitBeforeDeactivating){}”`
 - c. Add the InitializeThisTutorialTip function and inside of it, add listeners for the events that could activate or deactivate the tutorial tip. Those listeners should call CheckIfTipShouldBeActivated and CheckIfTipShouldBeDeactivated, respectively.
 - d. Inside of the CheckIfTipShouldBeActivated() and CheckIfTipShouldBeDeactivated() functions, check if the conditions for activating or deactivating the tutorial tip have been met. If they are met, call ActivateTutorialTip() or DeactivateTutorialTip().

HOW TO ADD A TUTORIAL TIP (EXAMPLE)

This is an example of what a tutorial tip's class should look like, from the tutorial tip that tells you "only buildings connected by roads make money." It only displays the tip if the player hasn't connected any buildings by roads. Note that it uses an extra function, `HousesAreConnected()` to determine this.

```
You, last month | 1 author (You)
1  using UnityEngine;
2  using System.Collections.Generic;
3  using System.Collections;
4
5  //"Houses Connected by roads make money" tutorial tip
6  public class ConnectHousesTutorialTip : TutorialTip
7  {
8      //Constructor passes values to base class
9      public ConnectHousesTutorialTip(int _tutorialTipTextID, TutorialTipManager _TTM,
10         float _timeToWaitBeforeActivating, float _timeToWaitBeforeDeactivating)
11         : base(_tutorialTipTextID, _TTM, _timeToWaitBeforeActivating, _timeToWaitBeforeDeactivating){
12     }
13
14
15     public override void InitializeThisTutorialTip(){
16         //Checks if houses are connected every time a tile is placed
17         GameManager.current.GetEvent(EventType.E.TileJustPlaced).AddListener(CheckIfTipShouldBeDeactivated);
18
19         //Checks if the tip should be activated once the player has placed a tile
20         GameManager.current.GetEvent(EventType.E.TileJustPlaced).AddListener(CheckIfTipShouldBeActivated);
21     }
22
23     public void CheckIfTipShouldBeActivated(){
24         if(!HousesAreConnected()){
25             ActivateTutorialTip();
26         }
27     }
28
29     //Checks if the Tip should be deactivated. Runs every time a tile is placed
30     public void CheckIfTipShouldBeDeactivated(){
31         if(HousesAreConnected()){
32             DeactivateTutorialTip();
33         }
34     }
35
36     //Checks if any houses are connected
37     private bool HousesAreConnected(){
38         //Checks if any of the houses are connected by roads
39         bool houseIsConnected = false;
40         foreach(Tile tile in TileTypeCounter.current.ResidenceTileTracker.GetAllTiles()){
41             if(tile is ActivatableTile activatableTile){
42                 if(activatableTile.IsActivated){
43                     houseIsConnected = true;
44                 }
45             }
46         }
47         return houseIsConnected;
48     }
49 }
50 }
```

INTERNAL API

These are some important public functions on a few scripts in the tutorial tip system.

TutorialTipManager.cs

void ActivateTutorialTip(int tutorialTipTextID)

This adds the tutorial tip that corresponds to the arbitrarily assigned tutorialTipTextID to the tutorial tip manager's list of active tutorial tips. It will display the tutorial tip if it's the one on top.

void DeactivateTutorialTip(int tutorialTipTextID)

This removes the tutorial tip that corresponds to the arbitrarily assigned tutorialTipTextID from the tutorial tip manager's list of active tutorial tips. It will then display whichever tutorial tip is on top of the list, deactivating them all if there are none in the list.

void DeactivateTopTutorialTip()

Deactivates whichever tutorial tip is currently being displayed. This is called when the player clicks the "X" to close the active tutorial tip.

TutorialTip.cs

void ActivateTutorialTip()

Waits however long should be waited before activating the given tutorial tip and then tells the tutorial tip manager to activate it, if its deactivation condition hasn't been met in the time it waited.

void DeactivateTutorialTip()

Waits however long should be waited to deactivate the tutorial tip then tells the tutorial tip manager to deactivate it.

TutorialTipGraphicsActivationManager.cs

void ActivateTutorialTipBackground()

Fades in the lightbulb graphic that sits behind the tutorial tip text.

void DeactivateTutorialTipBackground()

Fades out the lightbulb graphic that sits behind the tutorial tip text.